



SOMAIYA
VIDYAVIHAR UNIVERSITY

20.12.2022 (E)

Semester: August 2022 – December 2022		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 116U01	Class: TY	Semester:V (SVU 2020)
Programme: B.Tech Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP
Course Code: 116UC501	Name of the Course: Software Engineering	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Explain size oriented software product metrics.	5
ii)	State and define non-functional requirement in software engineering.	5
iii)	Describe refinement in software design process.	5
iv)	Describe the importance of deployment diagram in system implementation.	5
v)	State and define the goals of software testing.	5
vi)	Explain the different elements of activity diagram.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain the importance of estimation in software project planning.	5
ii)	What is Scrum? Explain its features.	5
	OR	
Q2 A	Explain the waterfall software development life cycle model and state its advantages and disadvantages.	10
Q 2 B	Solve any One	10
i)	Draw the use case diagram for library management system and state the function of extend and include relationship.	10
ii)	Explain the different types of relationship in class diagram with suitable example.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain user interface design rules in detail.	10
ii)	Define coupling and cohesion. Discuss the various types of coupling and cohesion.	10
iii)	Explain version control and change control in software configuration management.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	
i)	Explain three refactoring types in mapping model to code with suitable code for required part of the example.	20
ii)	State the different sources of risk and describe risk management approach.	10
iii)	Explain software maintenance. Describe different types of software maintenance.	10

Que. No.	Question	Max. Marks
Q5	Write notes on any four	
i)	State chart diagram	20
ii)	Component based software engineering	5
iii)	Component diagram	5
iv)	COCOMO Model	5
v)	Boundary value analysis	5
vi)	PERT for Risk analysis and management	5

22.12.2022(E)



Semester: August 2022 – December 2022		
Maximum Marks:	Examination: ESE Examination	Duration:3 hrs
Programme code: 03		
Programme: B Tech Computer Engineering	Class: TY	Semester: V (SVU 2020)
Name of the Constituent College:	Name of the department: COMP	
K. J. Somaiya College of Engineering		
Course Code: 116U01E511-	Name of the Course: Computer Graphics	
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary		

Question No.		Max. Marks
Q1 (a)	Explain application of Computer Graphics in Health Care.	05
Q1 (b)	Compare Random Scan and Raster scan display.	05
Q1 (c)	Write an algorithm to draw line using Bresenham's algorithm. OR Explain the OpenGL Line Functions citing examples with pseudo code.	10
Q2 (a)	Find the transformation matrix that transforms a square ABCD to half of its size with the center remaining at same position. The co-ordinates of the square are A(1,1), B(3,1), C(3,3), D(1,3) and center at (2,2). Also find the resultant co-ordinates of the square.	10
Q2 (b)	Discuss the fixed point 2-D scaling with an example. OR Describe the 3D viewing pipeline.	10
Q3 (a)	Use the Cohen Sutherland line clipping algorithm to clip two lines P1P2 and P3P4. Given data: P1(40,15)-P2(75,45) and P3(70,20) - P4(100,10) against a window A(50,10), B(80,10), C(80,40), D(50,40).	10
Q3 (b)	Explain 3 D rotation w.r.t all 3 co-ordinate axis. OR Explain Sutherland Hodgeman Polygon clipping with appropriate example.	10
Q4 (a)	Write in brief about Projections with diagram.	10
Q4 (b)	Discuss the OpenGL visibility detection functions. and give pseudo code. OR Write a short note on Depth Buffer Method with diagram.	10
Q5 (a)	Discuss Computer Animation and stages of Designing of Animation Sequences	10
Q5 (b)	Attempt any two: State the Bezier Spline Curves Properties. (5) Write short note on Color Models. (5) Explain the OpenGL curve functions. (5)	10

22-12-2022 (E)


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Semester: August 2022 – December 2022		
Maximum Marks: 100	Examination: ESE Examination	Duration: 3 Hrs.
Programme code: 01	Class: TY	Semester: V (SVU 2020)
Programme: B.Tech		
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering	
Course Code: J16U01E512	Name of the Course: Advanced Databases and Data Warehousing	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Draw the different parallel database architecture	5
ii)	Explain bitemporal databases with example	5
iii)	Briefly explain Features of NOSQL	5
iv)	What are the advantages of distributed database	5
v)	What are the characteristics of Data warehouse explain	5
vi)	List the advantages of OLAP w.r.t OLTP	5

Que. No.	Question	Max. Marks																		
Q2 A	Solve the following	10																		
i)	Give the difference between Inter Query and Intra Query Parallelism with an example	5																		
ii)	Compare Intraoperation Parallelism and Interoperation Parallelism with an example	5																		
OR																				
Q2 A	Explain the transparency features of a DDBMS.	10																		
Q 2 B	Solve any One	10																		
i)	Consider the following two sites in distributed database system: Site 1 :Database : Employee Number of Records : 100 <table><tr><th>Emp id</th><th>Name</th><th>Address</th><th>Contact No.</th><th>Department</th></tr><tr><td>4 bytes</td><td>8 bytes</td><td>20 bytes</td><td>5 bytes</td><td>4 bytes</td></tr></table> Site 2 :Database: Project Allotted , Number of Records : 80 <table><tr><th>Project id</th><th>Emp id</th><th>Project Name</th><th>Location</th></tr><tr><td>4 bytes</td><td>4 bytes</td><td>8 bytes</td><td>8 bytes</td></tr></table> Consider the following query. Q: "Find Name of the Employees working on the projects located "Delhi" OR "Mumbai" A. Show the strategies of decomposing and executing the above query at site 1 along with data transfer cost using Natural Join B. Find the data transfer cost using Semi Join Operation for the optimized strategy	Emp id	Name	Address	Contact No.	Department	4 bytes	8 bytes	20 bytes	5 bytes	4 bytes	Project id	Emp id	Project Name	Location	4 bytes	4 bytes	8 bytes	8 bytes	10
Emp id	Name	Address	Contact No.	Department																
4 bytes	8 bytes	20 bytes	5 bytes	4 bytes																
Project id	Emp id	Project Name	Location																	
4 bytes	4 bytes	8 bytes	8 bytes																	

ii)	What are the advantages, applications of spatial database. List the different data types available in spatial database.	10
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	<i>Compare SQL & NOSQL</i> databases. What are the advantages of using NoSQL explain with example	10
ii)	Explain Document type NOSQL and CRUD operation for managing library books	10
iii)	Explain Key value type NOSQL its features for storing products on Amazon	10

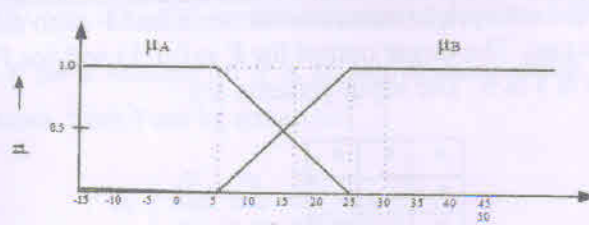
Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Explain the Data warehouse architecture with diagram	10
ii)	What are the different ways of Extracting data for Dataware	10
iii)	Why transformation of data required .Show the different types of transformation which can be applied on facebook users datawarehouse	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	<i>Explain primary copy method of Concurrently Control in distributed database</i>	5
ii)	Difference between Star schema and snowflake schēma	5
iii)	College wants to manages applicants all data w.r.t admissions. Suggest an analysis and an NOSQL database to store data, justify your suggestion	5
		5
v)	Compare ROLAP and MOLAP with architecture dagram	5
vi)	Give example for Slice and dice OLAP operation	5
(iv)	<i>List advantage and disadvantage of topdown approach of building datawarehouse</i>	

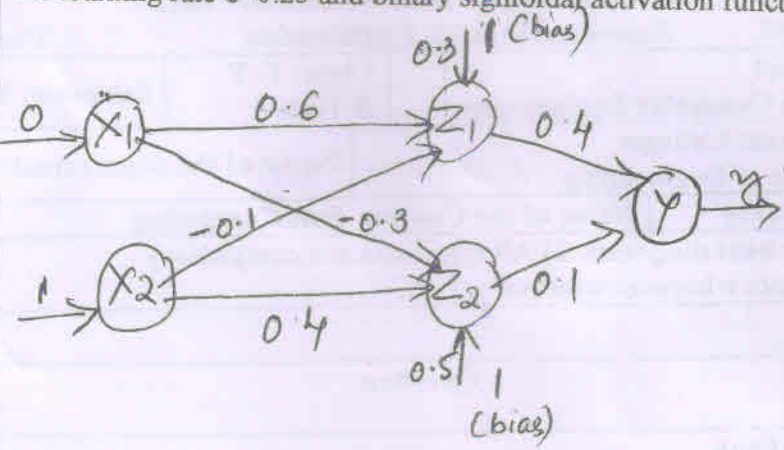
22.12.2022 (E)


SOMAIYA
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Semester: August 2022 – December 2022		
Maximum Marks: 100	Examination: ESE Examination	Duration: 3 Hrs.
Programme code: 1601	Class: T. Y	Semester: V (SVU 2020)
Programme: B. Tech Computer Engineering	B.Tech	
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116U01E514	Name of the Course: Soft Computing	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Draw and explain the McCulloch-Pitts neuron model	5
ii)	Explain windrow-Hoff learning rule.	5
iii)	State and explain Hebb rule used in training pattern association network	5
iv)	Write a short note on neocognitron model.	5
v)	 Consider two fuzzy sets A and B with membership functions $\mu_A(x)$ and $\mu_B(x)$, respectively defined as above. A = Cold climate with $\mu_A(x)$ as the MF. B = Hot climate with $\mu_B(x)$ as the M.F. Here, X being the universe of discourse representing entire range of temperatures. Plot the fuzzy set representing pleasant climate, extreme climate.	5
vi)	Define defuzzification. Explain First of maxima method with an example.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Differentiate between supervised learning and unsupervised learning	5
ii)	Train the autoassociative network for the input vector $[-1 \ 1 \ 1 \ 1]$ and also test the network with one mistake in test vector.	5
OR		
Q2 A	Perform two training steps of the network using delta learning rule for $\lambda = 1$ and $c = 0.25$. Train the network using the following data pairs $(X_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, d_1 = -1)$ and $(X_2 = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}, d_2 = 1)$ The initial weight are $W_1 = [1 \ 0 \ 1]^T$	10

Q 2 B	Solve any One	10																														
i)	Using back propogation algorithm, find the new weights for the net shown below. It is presented with the input pattern [0 1] and target output is 1. Use the learning rate $c=0.25$ and binary sigmoidal activation function 	10																														
ii)	Construct and test a BAM network to associate letters E and F with simple bipolar input-output vectors. The target output for E is (-1,1) and for F is (1,1). The display matrix size is 5 X 3. The input patterns are <table border="1" data-bbox="771 1075 933 1265"> <tr><td>*</td><td>*</td><td>*</td></tr> <tr><td>*</td><td>-</td><td>-</td></tr> <tr><td>*</td><td>*</td><td>*</td></tr> <tr><td>*</td><td>-</td><td>-</td></tr> <tr><td>*</td><td>*</td><td>*</td></tr> </table> “E” <table border="1" data-bbox="771 1366 933 1556"> <tr><td>*</td><td>*</td><td>*</td></tr> <tr><td>*</td><td>*</td><td>*</td></tr> <tr><td>*</td><td>-</td><td>-</td></tr> <tr><td>*</td><td>-</td><td>-</td></tr> <tr><td>*</td><td>-</td><td>-</td></tr> </table> “F”	*	*	*	*	-	-	*	*	*	*	-	-	*	*	*	*	*	*	*	*	*	*	-	-	*	-	-	*	-	-	10
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Draw Hopfield network with four output nodes. Also explain training and testing algorithm.	10
ii)	Consider the problem of designing fuzzy controller for a train approaching or leaving a station, the input are distance from a train and speed of the train. The output is brake power used. - Define the linguistic variables (use 3 for both input and output) - Design the fuzzy membership set using appropriate membership function - Form the rule base	10

iii)	Explain linearly separable and non-linearly separable with suitable examples	10
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Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	What is self-organizing map? Explain the training algorithm of Kohonen self-organizing feature maps.	10
ii)	Explain Center of sum and Center of gravity defuzzification methods with suitable example.	10
iii)	Compare different learning rules	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Explain winner takes all learning rule	5
ii)	Explain the principle behind simulated annealing network	5
iii)	Consider the following two sets P and D, which represent a set of paddy plants and a set of plant diseases $P = \{P_1, P_2, P_3, P_4\}$ a set of four varieties of paddy plants $D = \{D_1, D_2, D_3, D_4\}$ of the four various diseases affecting the plants. Let $S = \{S_1, S_2, S_3, S_4\}$ be the common symptoms of the diseases. Let, R be a relation on $P \times D$, representing which plant is susceptible to which diseases, then R can be stated as $R = \begin{matrix} & \begin{matrix} D_1 & D_2 & D_3 & D_4 \end{matrix} \\ \begin{matrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{matrix} & \begin{bmatrix} 0.6 & 0.6 & 0.9 & 0.8 \\ 0.1 & 0.2 & 0.9 & 0.8 \\ 0.9 & 0.3 & 0.4 & 0.8 \\ 0.9 & 0.8 & 0.4 & 0.2 \end{bmatrix} \end{matrix}$ Also consider T be the another relation on $D \times S$ which is given by $T = \begin{matrix} & \begin{matrix} S_1 & S_2 & S_3 & S_4 \end{matrix} \\ \begin{matrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{matrix} & \begin{bmatrix} 0.1 & 0.2 & 0.7 & 0.9 \\ 1.0 & 1.0 & 0.4 & 0.6 \\ 0.0 & 0.0 & 0.5 & 0.9 \\ 0.9 & 1.0 & 0.8 & 0.2 \end{bmatrix} \end{matrix}$ a) Obtain the association of plants with the different symptoms of the disease using max-min composition b) Also obtain $R \times T$	5
iv)	Compare and contrast BAM and Hopfield network	5
v)	What are the factors that can improve the convergence of learning in Back Propagation network.	5
vi)	Write a short notes on components of fuzzy inference system	5

24.12.2022(E)


SOMAIYA
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Semester: August 2022 – December 2022		
Maximum Marks: 100	Examination: ESE Examination	Duration: 3 Hrs.
Programme code: 01		
Programme: UG B.Tech in Computer Engg	Class: TY	Semester: V (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer	
Course Code: 116U01C502	Name of the Course: Computer Networks	
Instructions: 1) Draw neat diagrams. 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Discuss any 5 design issues for OSI layered architecture of data communication.	5
ii)	Explain following Link layer addressing scheme with example in IPv6 addressing: (Any One) a. Mapping EUI-64 b. Ethernet MAC address	5
iii)	Explain Count-to-infinity problem in Distance Vector Algorithm and one solution for the same.	5
iv)	Draw general frame structure of ICMP messages and explain Redirection message with help of diagram.	5
v)	Explain Leaky Bucket algorithm and Token Bucket algorithm.	5
vi)	Differentiate between Distance Vector and Link state Routing Algorithms (5 Points)	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain all fields of IEEE 802.3 Frame Structure with the help of neat diagram	5
ii)	Discuss about Throughput and vulnerable time of slotted Aloha with help of neat diagram.	5
	OR	
Q2 A	With the help of following terms, explain working of CSMA/CA: a. Interframe Spacing b. Contention Window c. Acknowledgements	10
Q2 B	Solve any One	10
i)	Explain working of Selective Repeat ARQ.	10
ii)	Explain CDMA with the help of an example and discuss use of Walsh table.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	What is DVMRP. Explain step wise working of DVMRP protocol.	10
ii)	Consider sending a 2400-byte datagram into a link that has a MTU of 700 bytes. Suppose the original datagram is stamped with identification number 422. How many fragments are generated? Draw the IP header for all the fragments generated with the values in the various fields in the IP datagram(s) generated related to the fragmentation.	10
iii)	A host A with IP address 130.23.3.20 and physical address AEBD12345678 has a packet to send to another host B with IP address 130.23.43.25 and physical address A1B2C3D46789. The two hosts are on the same network. Show the ARP request and reply packets encapsulated in ethernet frames.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Explain UDP checksum Calculation with the help of an example.	10
ii)	Explain Congestion control in TCP.	10
iii)	Explain state transition diagram of TCP.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	HTTP	5
ii)	TCP Timers	5
iii)	DNS	5
iv)	DHCP	5
v)	NAT	5
vi)	Telnet	5

24.12.2022(E)


SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: August 2022 – December 2022		
Maximum Marks: 100	Examination: ESE Examination	Duration: 3 Hrs.
Programme code: 01	Class: TY	Semester: V (SVU 2020)
Programme: UG B.Tech in Computer Engg		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116U01C502	Name of the Course: Computer Networks	
Instructions: 1) Draw neat diagrams. 2) All questions are compulsory		
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Que. No.	Question	Max. Marks
Q1	Solve any Four	20
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ii)	Explain following Link layer addressing scheme with example in IPv6 addressing: (Any One) a. Mapping EUI-64 b. Ethernet MAC address	5
iii)	Explain Count-to-infinity problem in Distance Vector Algorithm and one solution for the same.	5
iv)	Draw general frame structure of ICMP messages and explain Redirection message with help of diagram.	5
v)	Explain Leaky Bucket algorithm and Token Bucket algorithm.	5
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Q4	Solve any Two	20
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ii)	Explain Congestion control in TCP.	10
iii)	Explain state transition diagram of TCP.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	HTTP	5
ii)	TCP Timers	5
iii)	DNS	5
iv)	DHCP	5
v)	NAT	5
vi)	Telnet	5

27-12-2022 (E)


SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: August 2022 – December 2022		Duration: 3hours
Maximum Marks: 100	Examination: ESE Examination	
Programme code: 01	Class: FY/SY/TY/LY	Semester: I/II/III/IV/V/VI/VII/VIII (SVU 2020)
Programme: B Tech Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP/ETRX/EXTC/IT/MECH
Course Code: 116U01C503	Name of the Course: Operating System	
Instructions: 1) Draw neat diagrams 2) Assume suitable data if necessary		

Questi on No.		Max. Marks																								
Q1 (a)	Differentiate between monolithic and microkernel	05																								
Q1 (b)	Define the term Critical section and Race condition.	05																								
Q1 (c)	What is Process Control Block (PCB)?	05																								
Q1 (d)	Explain the effect of page size on the performance of a process.	05																								
Q2 (a)	Given five memory partitions of 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 212 KB, 417 KB, 112 KB, and 426 KB (in order)? Which algorithm makes the most efficient use of memory?	10																								
Q2 (b)	What are System calls? List and explain file-handling system calls. OR Explain Process control fork(), exec(), and wait() System calls.	10																								
Q3 (a)	Explain the following in brief:(anyone) 1. Semaphores 2. Scheduling in Linux system	10																								
Q3 (b)	Assume the following processes arrive for execution at the indicated time and the CPU burst time is given in ms. <table border="1"><thead><tr><th>Process</th><th>Burst Time</th><th>Arrival Time</th><th>Priority</th></tr></thead><tbody><tr><td>P1</td><td>10</td><td>0</td><td>5</td></tr><tr><td>P2</td><td>6</td><td>0</td><td>2</td></tr><tr><td>P3</td><td>7</td><td>1</td><td>4</td></tr><tr><td>P4</td><td>4</td><td>1</td><td>1</td></tr><tr><td>P5</td><td>5</td><td>2</td><td>3</td></tr></tbody></table> Find the Average waiting time, and the average turnaround time for FCFS, SJF(Non-preemptive), and Round Robin(Quantum-3ms)	Process	Burst Time	Arrival Time	Priority	P1	10	0	5	P2	6	0	2	P3	7	1	4	P4	4	1	1	P5	5	2	3	10
Process	Burst Time	Arrival Time	Priority																							
P1	10	0	5																							
P2	6	0	2																							
P3	7	1	4																							
P4	4	1	1																							
P5	5	2	3																							

Q4 (a)	Explain the difference between Paging and Segmentation. OR Explain the hardware support for Paging.	10																																																																																										
Q4 (b)	On a disk with 1000 cylinders, numbers 0-999, compute the number of tracks the disk arm must move to satisfy all requests in the disk queue. Assume the last request received was at track 345 and the head is moving toward track 0. The queue in the FIFO order contains requests for the following tracks 123, 874, 692, 475, 105, 376. Perform the computation for the following scheduling algorithms: i. FIFO ii. SSTF iii. SCAN	10																																																																																										
Q5 (a)	Consider the given snap of the System <table border="1"><thead><tr><th rowspan="2"></th><th colspan="4">Allocation</th><th colspan="4">Max</th><th colspan="4">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr></thead><tbody><tr><td>P0</td><td>0</td><td>2</td><td>1</td><td>2</td><td>0</td><td>3</td><td>2</td><td>2</td><td>2</td><td>5</td><td>3</td><td>2</td></tr><tr><td>P1</td><td>1</td><td>1</td><td>0</td><td>2</td><td>2</td><td>7</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P2</td><td>2</td><td>2</td><td>5</td><td>4</td><td>2</td><td>3</td><td>7</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td>0</td><td>3</td><td>1</td><td>2</td><td>1</td><td>6</td><td>4</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td>2</td><td>4</td><td>1</td><td>4</td><td>3</td><td>6</td><td>5</td><td>8</td><td></td><td></td><td></td><td></td></tr></tbody></table> Answer the following questions using Bankers algorithm. 1) What is the content of Matrix need? 2) Is the system in a safe state? 3) If a request from process P1 arrives for (1,3,2,1) can the request be granted immediately?		Allocation				Max				Available				A	B	C	D	A	B	C	D	A	B	C	D	P0	0	2	1	2	0	3	2	2	2	5	3	2	P1	1	1	0	2	2	7	5	2					P2	2	2	5	4	2	3	7	6					P3	0	3	1	2	1	6	4	2					P4	2	4	1	4	3	6	5	8					10
	Allocation				Max				Available																																																																																			
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P4	2	4	1	4	3	6	5	8																																																																																				
Q5 (b)	Explain the conditions for deadlock. Suggest techniques to avoid deadlock. OR Explain an algorithm for the producer-consumer problem.	10																																																																																										